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The S.An.A.® concept: Semi-aerobic, Anaerobic, Aerated bioreactor landfill

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ABSTRACT

Hybrid Bioreactor Landfills are designed to enhance and speed up biological processes, aiming at reducing the duration of post operational phase until landfill completion. S.An.A.® (Semi-aerobic, Anaerobic, Aerated) concept consists in a Hybrid Bioreactor featuring a first semi-aerobic phase to enhance the methane production occurring in the following anaerobic step and a forced aeration for the abatement of the residual emissions. At the end of the last step, semi-aerobic conditions are restored and flushing applied for leaching residual non-biodegradable compounds. Results of the application of S.An.A.® concept to a lab scale bioreactor system showed that pre-aeration was effective in controlling the concentration of VFA, increasing pH and stimulating methane production during anaerobic phase; in particular with intermittent airflow the methane potential was 50% higher respect to control reactors. Forced aeration reduced organic compounds and nitrogen concentration in leachate of an order of magnitude, better performing in low airflow reactors. S.An.A.® Hybrid bioreactors proved to be an efficient system both for increasing methane production and reaching landfill completion in shorter time, suggesting that with proper landfill management, the duration of post-closure care might be reduced by 25–35%.

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1. Introduction

Landfilling constitutes an unavoidable final step in waste management, being aimed at returning residues to the non-mobile state they were in prior to extraction from the ground and use as raw materials (Cossu, 2016). Although European policies envisage a consistent reduction of untreated MSW deposition (EU, 2015), the final disposal of waste will remain a fundamental tool for closing the material cycle. "A Circular Economy cannot exist in the absence of a sustainable closure of the material loop" (Cossu, 2016).

In line with the European sustainability principle (EU, 2006), reduction of the duration of post operational phase towards landfill completion should be considered a priority and a driver in the proper management and design of landfills. This is currently a heavily debated topic in scientific literature (Laner et al., 2012; Ritzkowski et al., 2016), with local authorities, such as the Lombardy Region (Italy), starting to include this concept in their guidelines (Deliberation of the Regional Council 2461/14). Some authors report that landfills can be safely closed by enhancing the biochemical processes occurring inside and thus guaranteeing environmental acceptability of future emissions (Christensen et al., 2011; Ritzkowski and Stegmann, 2013). To this regard, the Bioreactor Landfill concept represents a valuable tool for use in reducing the potentially lengthy periods required to achieve these conditions.

Bioreactor Landfills are designed to accelerate waste stabilization processes and several examples of lab and full-scale applications are available (Price et al., 2003; Xu et al., 2014; Berge et al., 2009; Townsend et al., 2015; Cossu et al., 2016). Bioreactors are defined by use of a series of technologies including water or wastewater injection (Cossu et al., 2003; EPA, 2015; Morello et al., 2016), forced or natural aeration (Ritzkowski et al., 2016; Price et al., 2003; Sekman et al., 2011; Raga et al., 2015), leachate recirculation (Sandip et al., 2012; Morello et al., 2016) and other combinations of in situ treatments (Townsend et al., 2015). These options facilitate the control of pH, redox conditions, moisture content, distribution of nutrients and microorganisms (Norbu et al., 2005), oxygen supply for nitrification process, and leaching of inorganics, among others (Shao et al., 2008; He et al., 2011; Erses et al.,

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2008; Xu et al., 2014). Three main types of landfill bioreactors are commonly considered as far as the metabolic pathway is concerned: anaerobic, aerobic and hybrid bioreactors, the latter providing for a sequence of aerobic and anaerobic conditions (EPA, 2015; Townsend et al., 2015). In particular, Hybrid Bioreactors operate under a series of combinations of aerobic and anaerobic conditions to achieve benefits from both, potentially enhancing biochemical processes and allowing methane production as well as nitrification-denitrification of ammonium ions (Xu et al., 2014; Townsend et al., 2015).

This paper presents an application of the S.An.A.® (Semi-aerobic, Anaerobic, Aerated) Landfill concept (Repetti et al., 2013), which consists in a hybrid bioreactor featuring an initial semi-aerobic phase to enhance methane production occurring in the subsequent anaerobic step and, finally, forced aeration for the abatement of residual emissions (Cossu et al., 2016). At the end of the last step, naturally occurring semi-aerobic conditions will be restored and flushing can be applied for the leaching of residual non-biodegradable compounds.

The aim of the present study was to apply the S.An.A.® concept to four lab-scale landfill simulation reactors, analyzing different management options and comparing final results with two traditional anaerobic reactors. Phase 1 was designed to combine the demonstrated advantages of leachate recirculation and landfill pre-aeration, with a greater methane production monitored during Phase 2. Preliminary results were presented in Cossu et al., 2016. This paper reports the results of the full experiment in which the anaerobic Phase 2 was followed by forced aeration in Phase 3, with the aim of completing biological stabilization processes and a final optional Phase 4 involving flushing of inorganics. Flushing is an efficient technology, able to leach out the soluble fractions of landfill contaminants until reaching low concentrations (Christensen et al., 2011). However, this technology is generally too expensive and holds some severe operational challenges, which may prevent the measure from being applied more often in real landfills (Bolyard and Reinhart, 2016). Among the others, introduction of clean water in landfills is forbidden by law or not authorized by public authorities, which consider this practice an unsustainable contamination of clean liquids (Morello et al., 2016).

In this paper, the results of the entire test were taken into consideration; in particular, final quality of waste, degradation kinetics and comparison between reactor performances were analyzed.

2. Materials and methods

2.1. Waste samples

Waste used for the lab-scale test was the residual fraction of Municipal Solid Waste after source segregation, provided by a public waste management company operating in thirteen municipalities of Livorno Province, Tuscany (Italy). A 200 kg sample was sieved using an 80 mm mesh and the fractional composition analyzed (Table 1); the undersieve (<80 mm fraction) was loaded into lab-scale reactors (Cossu et al., 2016).

Table 1
Fractional composition of raw waste and undersieve (<80 mm), i.e. the fraction used to fill the reactors. Modified from Cossu et al. (2016).

Trade class	Raw waste (%)	Undersieve <80 mm (%)
Food waste	14.0	17.4
Green waste	6.1	6.0
Paper	22.9	10.5
Plastic	18.4	12.6
Metal	5.2	6.2
Glass and inert	3.5	3.2
Wood	0.4	0.3
Textiles	8.5	5.1
Undersieve <20 mm	20.9	38.8

2.2. Equipment

The experiments were carried out using six Plexiglass® (polymethyl methacrylate) columns (height 106 cm, diameter 24 cm), each filled with 18.4 kg MSW. A 10-cm thick gravel layer (Ø 20–40 mm) was placed at the bottom of each reactor as a drainage layer to facilitate the extraction of leachate. Another similar gravel layer was placed at the top of each column to increase the distribution of distilled water and recirculated leachate (Fig. 1).

The upper end of the reactor body was equipped with three valves providing for the introduction of air, sampling and extraction of gas as well as introduction of liquids. A leachate collection port was located at the bottom of each column and leachate extracted fell directly into a collection tank (5 L capacity). Recirculation of leachate was carried out using programmable peristaltic pumps (Heidolph PD 5001), which automatically pump up leachate from the tank to the liquids introduction valve placed in the top of the reactor. The leachate collection and recirculation system was built to avoid any leakage of leachate and any loss of gas in the circuit (Fig. 1).

To channel air into the waste body, a vertical PVC pipe with side perforations was installed at the center of the waste layer; this system was designed to guarantee the uniform distribution of air throughout the reactor. Air injection was provided by Prodac Air Professional pump 360, and airflow was regulated by a Sho-Rate GT1135 flow meter. Biogas generated from each column was collected by a Tedlar® sampling bag connected to the upper gas port and its volume and quality were measured by means of a volumetric flow meter.

Temperature monitoring was carried out using Thermo Systems TS100 temperature probes installed inside the reactor and the temperature maintained constant at 38–40 °C by means of a thermo-regulated insulation system covering all lateral surfaces of the reactor (Fig. 1).

2.3. Test setup

The experiment was divided into three distinct main phases: Semi-aerobic, Anaerobic and Aerobic, in line with the S.An.A.® concept introduced by Repetti et al. (2013) and tested for the first two phases as described in Cossu et al. (2016). Phase 1 was intended to enhance biogas production in the subsequent anaerobic Phase 2. Phase 3 was functional to complete waste stabilization by means of forced aeration. Phase 4 with flushing under semi-aerobic conditions was aimed at removing residual salts and heavy metals (HM).

The six column reactors were filled with the same quantity of waste and run under identical hydraulic conditions for the whole test. Two reactors (ANa and ANb) were run under anaerobic conditions and used as control. Hybrid lab-scale reactors were managed under different conditions as represented in Fig. 2 and described below:

- *Reactor HF-I:* Intermittent semi aerobic Phase 1 and High Air Flow rate during forced aeration.
- *Reactor HF-C:* Continuous semi aerobic Phase 1 and High Air Flow rate during forced aeration.
- *Reactor LF-I:* Intermittent semi aerobic Phase 1 and Low Air Flow rate during forced aeration.
- *Reactor LF-C:* Continuous semi aerobic Phase 1 and Low Air Flow rate during forced aeration.

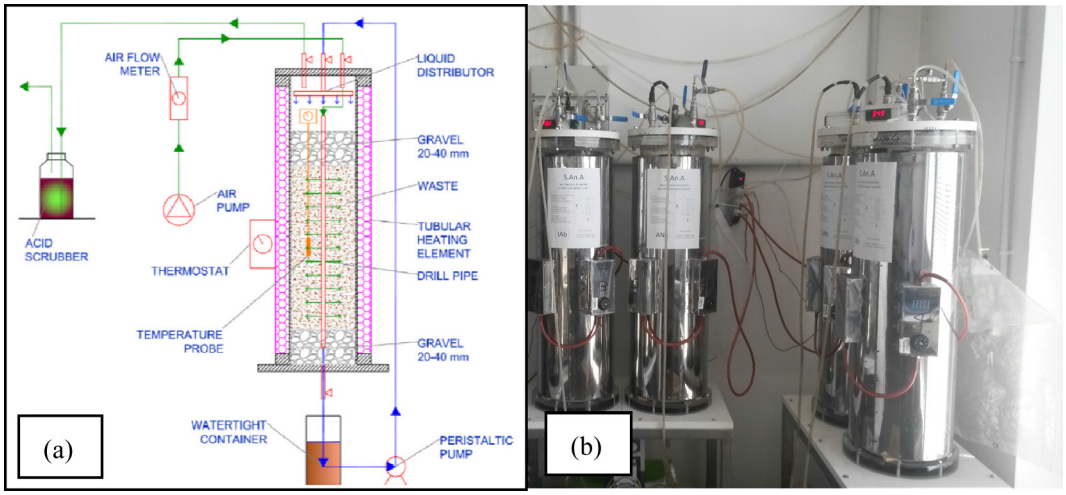


Fig. 1. Reactor setup: (a) sketch and (b) view.

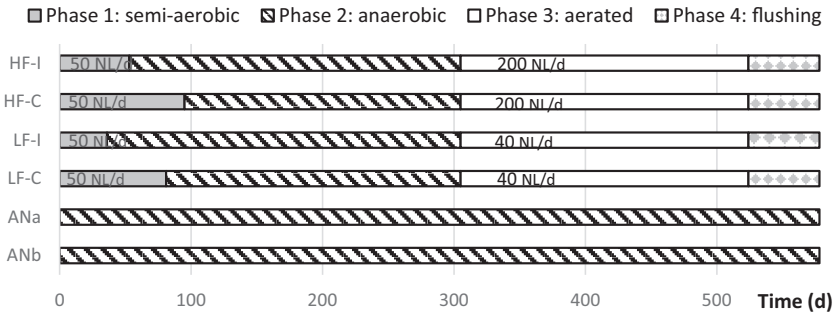


Fig. 2. Lab-scale reactor air management. AN control reactor operated solely in anaerobic conditions while S.An.A.® reactors were subjected to four different phases.

At the beginning of the test, 5 L of distilled water were introduced into each reactor and recirculated several times to saturate waste field capacity and produce 2–3 L of process water for sampling and recirculation. Leachate samples (0.1–0.25 L) were collected periodically for analysis and replaced each time with the same amount of distilled water.

After filling, aeration was commenced in hybrid reactors to simulate semi-aerobic conditions (Fig. 2). At test start up, airflow was set at a low regime and incremented until oxygen was detectable in the off gasses (O_2 > 1–2%). Daily air flow was the same for all pre-aerated reactors (50 L/d at 20 °C) which, in our specific case, corresponded to 5 NL/d/kgTS, TS being the dry matter content (Total Solids) in the reactor. As described by Cossu et al. (2016), this air flow was chosen on the basis of a series of lab-scale experiments carried out on landfill lab-scale reactors in semi-aerobic conditions (Cossu et al., 2003; Sun et al., 2013; Wu et al., 2014). Intermittent aeration (-I reactors) was turned on for 12 h a day (air flow approximately equal to 0.42 NL/h/kgTS), while continuous aeration (-C reactors) was provided 24 h a day, seven days per week (air flow equal to 0.21 NL/h/kgTS).

Transition between the different phases was carried out by means of a performance-based method consisting in continuous monitoring of specific parameters in leachate (Table 2). In particular, the transition between semi-aerobic Phase 1 and the subsequent anaerobic Phase 2 was conducted by monitoring specific chemical parameters capable of guaranteeing optimal conditions for methanogenic bacteria (Cossu et al., 2016).

Anaerobic Phase 2 lasted until biogas flux was no longer detectable for all the reactors (lower than 0.5 L_{CH4}/d); biogas quality was monitored constantly throughout the whole period and temperature maintained at 38–40 °C.

During subsequent Phase 3, air was channelled continuously into S.An.A.® reactors to simulate forced aeration and thus enhance biological degradation processes and promote nitrification for ammonium depletion (Townsend et al., 2015; Berge et al., 2006; Ritzkowski et al., 2016). In Low Air Flow (LF) reactors, air injection was calibrated to maintain oxygen in off gas detectable but lower than 4% (LF-I and LF-C). Consequently, air injection was 40 NL/d, corresponding to approximately 7.7 NL/kgTS/d (Fig. 2). In High Air Flow (HF) reactors, air injection was carried out with the purpose of achieving oxygen levels of more than 14% in off gas (HF-I and HF-C), thus limiting the risk of poorly aerated zones in the reactor (Ritzkowski and Stegmann, 2013; Morello et al., 2016). Air injection in HF reactors was 200 NL/d, corresponding to approximately 38.5 NL/kgTS/d (Fig. 2).

Forced aeration was stopped once waste and leachate characteristics in all S.An.A.® reactors featured values constantly below those illustrated in Table 2. Among them, BOD_5/COD ratio below 0.1, ammonium below 50 mg/L and respiration index in solid waste (RI_4) under 2.5 mgO_2/kgTS were recommended to reach biochemical stability of waste (Laner et al., 2012; Sekman et al., 2011; D.G.R. 2461/14).

After aeration stop (day 524), reactors were again managed under semi-aerobic conditions, with natural air circulation enhanced by opening the top and the bottom valves of reactors,

Table 2
Performance-based parameters and values for leachate samples used to determine the transition between phases and the end of the test.

	Parameter	Values range	References
Transition from Phase 1–2	pH	6.7–7.4	Lay et al. (1998)
		6.4–7.2	Chugh et al. (1998)
		6.5–8.2	Sekman et al. (2011)
		6.65–7.41	Sandip et al. (2012)
		6.25–8.00	Cossu et al. (2016)
	Alkalinity ($mgCaCO_3/L$)	1000–5000	Agdag and Sponza (2005)
		2000–3500	Sekman et al. (2011)
	Volatile fatty acids ($mgCH_3COOH/L$)	<6000	Wang et al. (1999)
		<7100	Cossu et al. (2016)
Transition from Phase 2–3	Area-specific methane production[a] ($L_{CH4}/d/m^2$)	<0.5	Detection Limit
		<0.1–0.5	Laner et al. (2012)
Stop of forced aeration	COD (mgO_2/L)	<1500	D.G.R. 2461/14
	BOD_5/COD	<0.1	D.G.R. 2461/14
		<0.13	Sekman et al. (2011)
	Ammonium (mgN/L)	<300	Laner et al. (2012)
		<50	D.G.R. 2461/14
	RI_4 (mgO_2/gTS)[b]	<2,5	Laner et al. (2012)
Test end	Nitric nitrogen (mgN/L)	<20	D.G.R. 2461/14
	Sulfates (mg/L)	<1000	D.G.R. 2461/14
	Chloride (mgN/L)	<1200	D.G.R. 2461/14

[a] Methane was monitored in biogas emissions.
[b] RI_4 was monitored in solid samples.

increasing water supply to simulate flushing (Fig. 2). Leachate production was increased to 1 L/week to stimulate the dilution and leaching of inorganic compounds (sulfates, chloride, nitric nitrogen and heavy metals). This Phase 4 lasted for 7 weeks, starting from L/S ratio = 2.19 L/kgTS, reaching 3.35 L/kgTS, when all target values illustrated in Table 2 were achieved.

At the end of the test, all monitored parameters in hybrid reactors reached values listed in Table 2; in the anaerobic control reactor ammonium leaching was not sufficient to reach a concentration lower than 50 mgN/L, even after flushing.

At the end of the test, monitoring data were used to evaluate total compound emissions for each column (carbon, nitrogen, chloride and HM), and estimate the quantity of compounds still present in the waste.

Reaction kinetics and leaching rates were evaluated to allow a mathematical comparison to be made between performances of S.An.A.® reactors and controls. Biogas production during Phase 2 was simulated by means of a kinetic model derived from the Gompertz Growth Equation, calibrated with biogas daily data (Cossu et al., 2016). This model has a sigmoid shape with a clear inflection point and its output is the cumulative volume of methane produced from the lab-scale reactor according to Eq. (1) (Lay et al., 1998; Sandip et al., 2012).

$$BG = BGP * \exp\left\{-\exp\left[\frac{Rm * e}{BGP} * (\lambda - t) + 1\right]\right\} \quad (1)$$

BG is cumulative methane yield (mL/gVS), t is cumulated time (d), BGP is methane yield potential (mL/gVS), R_m is maximal daily methane yield (mL/gVS/d), λ is bacteria growth lag time (d) and e is a mathematical constant (=2.718). Calibration of the Gompertz Equation was carried out by minimizing mean square deviation of the values measured and calculated.

COD and ammonium ion consumption during Phase 3 were simulated by means of a first order kinetic Eq. (2): C_t is concentration at time t, C_0 is the initial concentration and k (t^{-1}) the kinetic constant. Similarly, chloride leaching in Phases 3 and 4 was simulated with a first order equation calibrated on L/S ratio as control variable, instead of time.

$$C_t = C_0 * e^{-k*t} \quad (2)$$

Calibration of the single kinetic constant was carried out by minimizing mean square deviation of the values measured and calculated.

2.4. Analytical methods

International standard methods were used for the analysis of solid samples, leachate and biogas.

A portable analyser (Eco-Control LFG20) was used to measure CO_2, CH_4 and O_2 concentrations in biogas, collected in the Tedlar bags, on at least a weekly basis. Stripped $N\text{-}NH_{3(g)}$ was measured through an acid scrubber placed immediately after the off-gas valve of each reactor (Fig. 1). Boric acid 0.5 M was used as scrubber and emission results were periodically compared with a portable gas analyser (Analitica Strumenti LFG 2000).

Leachate samples were analyzed periodically for pH, COD, TOC, BOD_5, TKN, $N\text{-}NH_4^+$, $N\text{-}NO_3^-$, SO_4^{2-}, Cl^- and Heavy Metals (Cr, Cu, Fe, Mn, Ni, Pb, Zn) according to the prescribed Italian standards, derived from international certified procedures (CEN, 2002).

Waste was sampled for characterization prior to reactor filling and at the end of the test, following complete extraction of the material and mixing in a tank to increase homogeneity.

Waste characterization at the end of the anaerobic phase comprised complete extraction of the waste from each reactor and mixing in a tank to increase homogeneity. Following this procedure, solid samples were milled to 4 mm prior to determining TS, VS, TOC, TKN, Cl^-, RI_4 and heavy metals. TOC on solid samples was measured using a TOC-VCSN Shimadzu Analyzer. Respiration Index (RI_4 and RI_7 mgO_2/gTS) was determined by means of Sapromat apparatus (H + P Labortechnik, Germany). A leaching test was carried out according to the standard UNI EN 12457-2 on waste samples (L/S was brought to 10 L/kgTS, mixed for 24 h and filtered at 0.45 μm). The eluate obtained was analyzed to evaluate the emission potential of some contaminants considered in leachate (COD, BOD_5, $N\text{-}NO_3^-$, and Cl^-). BOD_5/COD ratio is a parameter capable of indicating the amount of biodegradable compounds still present in a liquid (Cossu et al., 2012), assuming values between 0.02 and 0.13 for low biodegradability (Sekman et al., 2011) and values exceeding 0.4 for high biodegradability (Kjeldsen et al., 2002).

All tests on solid samples were performed at least in duplicate.

Table 3
Solid waste characterization in initial and final samples (Average ± SD).

	Initial sample	Final samples					
	All reactors	HF-I	HF-C	LF-I	LF-C	ANa	ANb
TS (%)	55.0 ± 1.0	37.0 ± 0.1	40.9 ± 0.1	41.4 ± 0.5	43.2 ± 1.6	40.1 ± 2.7	40.9 ± 0.3
VS (%TS)	58.9 ± 1.1	36.1 ± 0.1	32.5 ± 1.6	28.4 ± 2.7	22.8 ± 2.4	44.0 ± 0.4	36.5 ± 0.1
TOC (gC/kgTS)	368 ± 10	217 ± 38	197 ± 44	220 ± 47	216 ± 20	221 ± 12	226 ± 39
TKN (gN/kgTS)	9.7 ± 0.5	7.1 ± 0.4	7.0 ± 0.5	7.2 ± 0.1	7.3 ± 0.1	7.6 ± 0.6	7.6 ± 0.2
Cl^- (gCl-/kgTS)[a]	7.2 ± 0.1	1.5 ± 0.1	1.1 ± 0.2	1.0 ± 0.1	1.2 ± 0.1	1.2 ± 0.1	1.1 ± 0.2
RI_4 (mgO_2/gTS)	76.9 ± 2.4	2.4 ± 0.2	2.1 ± 0.1	2.2 ± 0.2	2.3 ± 0.1	6.2 ± 0.8	6.0 ± 0.2
RI_7 (mgO_2/gTS)	79.3 ± 4.4	4.5 ± 0.3	4.3 ± 0.2	3.9 ± 0.3	3.8 ± 0.1	9.4 ± 1.3	9.4 ± 0.2
BOD_5/COD[a]	0.51 ± 0.2	0.04 ± 0.1	0.04 ± 0.1	0.06 ± 0.1	0.04 ± 0.1	0.04 ± 0.1	0.04 ± 0.1
N-NO_3^- (mg/L)[a]	<10	<1	<1	<1	<1	<1	<1

[a] Obtained thoughout analysis on eluate of a standard leaching test UNI EN 12457-2.

3. Results

3.1. Solids

The fractional composition of waste loaded into reactors was characterized by high amounts of food waste, plastic, paper and fines. Minor quantities of green waste, metals, glass, wood and textiles were detected (Table 1).

The results of waste characterization at both the beginning and the end of the test are reported in Table 3. Waste wet mass decreased from the initial value of 18.4 kg per column (Total Solids TS = 55.0%), to values between 12 and 13 kg in the different reactors at the end of the test (TS = 37.1–43.2), indicating a significant drop in waste dry mass. Concomitantly, Volatile Solids (VS) displayed a similar trend, starting from 58.9%TS and decreasing to 22.8–44.0%TS, due to degradation processes. TOC decreased by about 30% during Phases 1 and 2 (Cossu et al., 2016) and by about 60% throughout the whole test.

During the test, BOD_5/COD ratio decreased from an initial value of 0.51 to reach values below 0.06 in all reactors (Table 3). RI_4 and RI_7 Respiration Indices confirmed the low residual degradability of samples. From initial values of 76.9 mgO_2/gTS, RI_4 decreased to reach values of less than 2.5 mgO_2/gTS (Table 3), the target value indicated by Laner et al. (2012) in Table 2.

TKN content in solid samples decreased slightly in all columns during Phases 1 and 2 (Cossu et al., 2016) as nitrogen emissions occurred only via leaching and no nitrification was detected. On the contrary, during forced aeration in Phase 3, the presence of oxygen promoted the nitrification process, which consumed efficiently ammonia ion. TKN content in solid waste was reduced by 24% in S.An.A.® reactors, whilst only an 18% reduction was detected in anaerobic control reactors (Table 3).

Chloride emission potential decreased by 79–85% from the beginning to the end of the test due to rapid dissolution into leachate, as reported in other similar landfill simulation experiments (Fellner et al., 2009).

3.2. Leachate

Evolution of the main parameters of carbon, nitrogen, sulfates and chloride, monitored in leachate samples extracted from the reactors throughout the test, is reported in Figs. 3 and 4.

Effect of pre-aeration on pH increase due to VFA consumption is clearly visible in Fig. 5. As illustrated in Table 2, pre-aeration in S.An.A.® reactors was stopped after reaching a pH of 6.25 and 7100 $mgCH_3COOH$/L VFA, corresponding to 35 d (LF-I) and 55 d (HF-I) in intermittent reactors, and 81 d (LF-C) and 95 d (HF-C) in continuous ones. These test phases have been discussed in Cossu et al. (2016), reporting a beneficial effect of pre-aeration, especially with intermittent airflow (HF-I and LF-I).

At the beginning of Phase 2, methane production started and COD dropped from about 70,000 mg/L to less than 5000 mg/L and BOD_5/COD ratio decreased from 0.8 to 0.1 (Fig. 3), reaching biochemically-stable conditions (Sekman et al., 2011).

Total Nitrogen (TN) in leachate was initially composed mainly by organic nitrogen (500–700 mgN/L) and in less extent by ammonium nitrogen (300–400 mgN/L). During Phases 1 and 2, ammonification consumed organic nitrogen until concentrations under 200 mgN/L, while producing NH_4^+ which only possible emission was solubilization and leaching. As consequence, TN concentrations remained always higher than 1000–1500 mgN/L in all reactors. During Phase 3, forced aeration allowed nitrification processes, consuming the residual organic nitrogen and the ammonia ion until reaching the established target value of 50 mg/L (Fig. 3). As observed by other authors (Berge et al., 2006; Raga and Cossu, 2013) denitrification is likely to occur in anoxic zones of aerated reactors. During aerobic Phase 3, lower NO_3^- concentrations than those expected from stoichiometry calculations for nitrification were detected, suggesting the possible development of simultaneous denitrification processes. NO_3^- concentrations up to 300 mg/L were observed in HF reactors; lower values, always below 50 mg/L, were measured in LF reactors where the lower air flow probably caused the presence of more extensive anoxic zones suitable for denitrification than in HF reactors (Fig. 3).

Sulfate concentrations decreased constantly during Phases 1 and 2, from an initial 2250 mg/L, to 500 mg/L. During aeration in Phase 3, the opposite process happened due to high oxygen availability, with sulfate concentrations once again increasing in S.An.A.® reactors only (Fig. 4). Finally, in Phase 4, all remaining sulfates were removed by flushing and final concentrations below the target value of 1000 mg/L were reached.

Cl^- emissions were dependent only on solubilization and leaching. From initial values of up to 4000 mg/L, chloride concentration decreased to below the target value of 1200 mg/L in the same way in all reactors (Fig. 4).

3.3. Biogas

The evolution of cumulative gaseous emissions is reported in Fig. 5. Methane, carbon dioxide, oxygen and stripped ammonia were all closely monitored.

Methane emissions were limited to anaerobic Phase 2 and influenced by pre-aeration mode: intermitted reactors emitted up to 32 gC/kgTS of methane, performing better than continuous reactors and consistently better than control reactors, which produced less than 15 gC/kgTS of methane (Cossu et al., 2016).

Carbon dioxide emissions in hybrid reactors were registered throughout the test period, in concentrations up to 70% during Phase 1 and around 40% during anaerobic Phase 2; subsequently, these percentages progressively decreased to zero. The higher

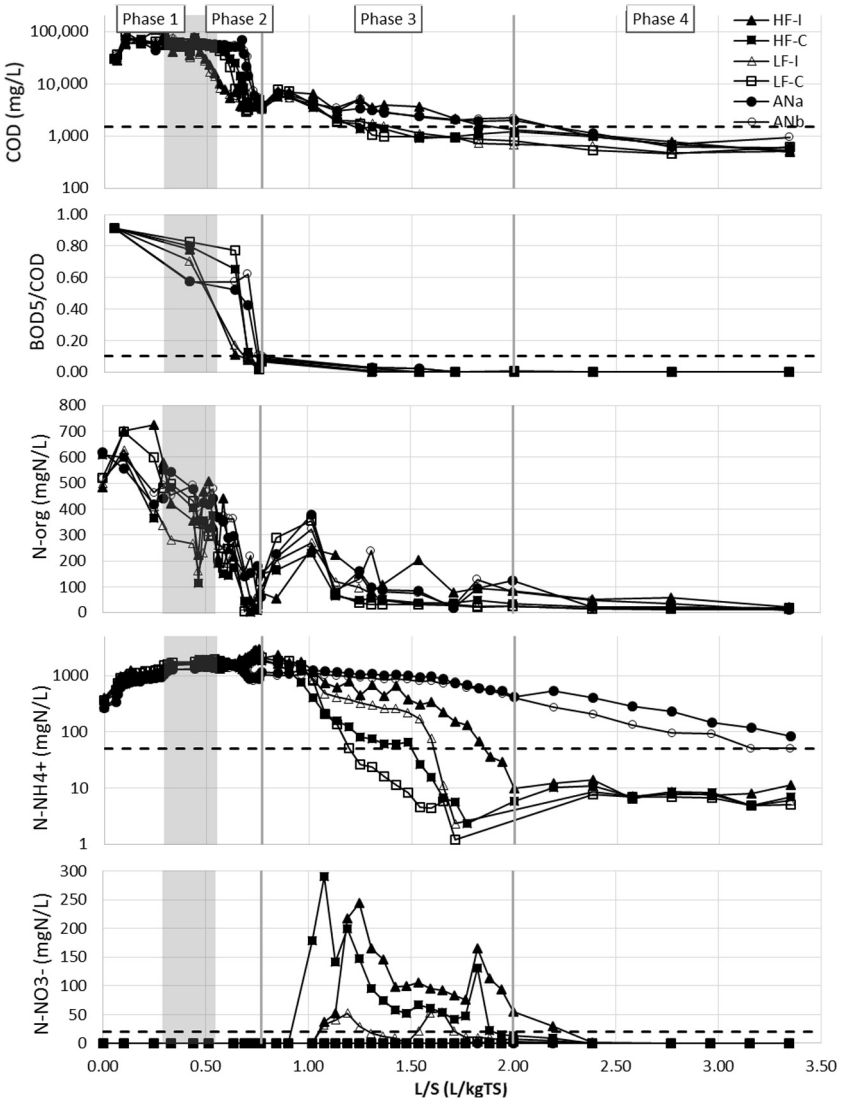


Fig. 3. Carbon and Nitrogen index monitoring in leachate (COD, BOD_5/COD, N-org, $N\text{-}NH_4^+$, $N\text{-}NO_3^-$). Grey vertical lines indicate the passage between the four phases while the dotted horizontal line indicates the performance-based limits of Table 2. COD and $N\text{-}NH_4^+$ are reported in logarithmic scale.

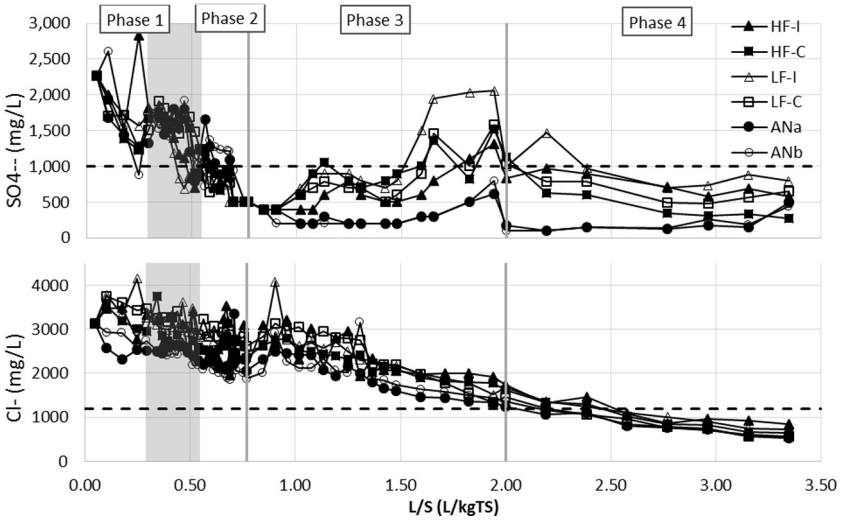


Fig. 4. Sulfates and chloride monitoring in leachate. Grey vertical lines indicate the passage between the four phases while the dotted horizontal line indicates the performance-based limits of Table 2.

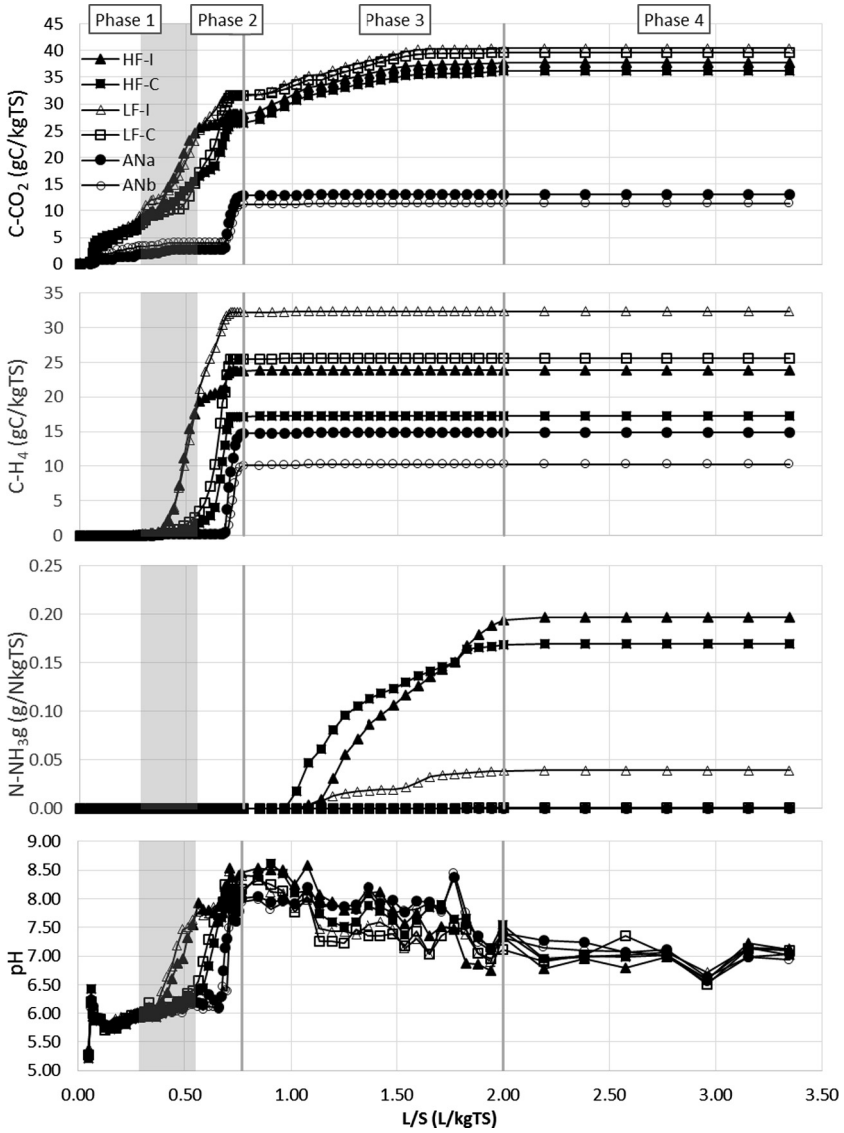


Fig. 5. Carbon dioxide, methane and stripped ammonia monitoring in gaseous emissions and pH monitoring in leachate. Grey vertical lines indicate the passage between the four phases.

carbon mass emissions in biogas were detected during Phase 2, concomitant with methane production, and during aeration in Phase 3 (Fig. 5). At the end of the test, S.An.A reactor emitted 35–40 gC/kgTS, while control columns emissions were below 12 gC/kgTS.

Stripped ammonia volatilization was only observed during Phase 3 in aerated reactors, although pH values were slightly below 8 (Fig. 5). HF reactor emitted a total of 0.17–0.20 gN/kgTS, LF-I reactor emitted 0.04 gN/kgTS, while no emissions were detected in LF-C and anaerobic reactors.

4. Discussion

4.1. Emissions evaluation

Total emissions percentages, respect to initial content of each compound in the waste sample, are reported in Table 4.

TOC emissions in leachate totalled 2.4–2.6% respect to initial content in solid samples in S.An.A.® reactors, and were a little higher (3.1%) in control ones in which the concentrations of organic compounds in leachate remained higher. Total methane production (5.4–7.7% of initial content in solid samples) was lower than carbon dioxide production (9.8–10.6% of initial content in solid samples) due to the forced aeration performed during Phase 3 (Table 4). These data also confirm that carbon emitted via leachate in aerated lab-scale reactors is generally one order of magnitude lower than carbon emitted in biogas; this finding is in agreement with the observations of other authors (Lornage et al., 2013). Monitoring of control reactors highlighted a fifty percent reduction in carbon gaseous emissions compared to hybrid reactors, confirming the effectiveness of pre-aeration in methane generation enhancement (Cossu et al., 2016; Sang et al., 2009; Erses et al., 2008), and the need for aerobic treatment to achieve a persistent depletion of organic compounds (Heyer et al., 2013; Raga and Cossu, 2013; Ritzkowski and Stegmann, 2013).

Nitrogen emissions in leachate were mainly due to ammonium ions and, to a lesser extent, to organic nitrogen, which had been ammonified almost completely during Phases 1 and 2, and nitrates, which had been produced during forced aeration in Phase 3 (He et al., 2011; Ritzkowski et al., 2006). Cumulative emissions of ammonium ions were found to be higher in control (22.6% of initial

Table 4
Percentages of carbon, nitrogen, chloride and heavy metal emissions respect to initial compound content in solid waste sample.

Emissions (%)			HF	LF	AN
Carbon	Leachate	TOC	2.6 ± 0.1	2.4 ± 0.4	3.1 ± 0.1
		VFA	0.3 ± 0.1	0.3 ± 0.1	0.4 ± 0.1
	Gas	$C\text{-}CH_4$	5.4 ± 1.2	7.7 ± 1.3	3.3 ± 0.9
		$C\text{-}CO_2$	9.8 ± 0.3	10.6 ± 0.1	3.2 ± 0.3
Nitrogen	Leachate	$N\text{-}NH_4^+$	16.9 ± 4.3	16.4 ± 0.4	22.6 ± 2.3
		N-Norg	5.0 ± 1.2	4.3 ± 0.6	5.6 ± 0.2
		$N\text{-}NO_3^-$	1.1 ± 0.1	0.2 ± 0.1	0.0 ± 0.0
	Gas	$N\text{-}NH_{3g}$	1.9 ± 0.2	0.3 ± 0.2	0.0 ± 0.0
Chloride	Leachate	Cl^-	88.6 ± 3.6	90.4 ± 2.2	73.8 ± 1.9
Heavy Metals	Leachate	Cr	0.7 ± 0.1	0.7 ± 0.1	0.7 ± 0.3
		Cu	2.9 ± 0.3	2.1 ± 0.2	1.6 ± 0.1
		Fe	1.3 ± 0.1	1.3 ± 0.2	1.2 ± 0.1
		Mn	7.8 ± 2.2	7.9 ± 2.4	10.0 ± 0.8
		Ni	6.5 ± 0.2	6.9 ± 2.0	8.5 ± 0.7
		Pb	1.2 ± 0.8	0.7 ± 0.1	0.8 ± 0.1
		Zn	4.4 ± 0.5	3.4 ± 0.4	3.6 ± 0.2

content in solid samples) than in S.An.A.® reactors (16.4–16.9%) because nitrification-denitrification was not possible in AN columns, and because the fact that high NH_4^+ concentrations in leachate persisted until the end of the test (Table 4). Nitrate emissions were monitored only in aerated reactors, particularly in HF (1.1% of initial content in solid samples) in which denitrification was slowed down by the excess of oxygen. Total Nitrogen emissions in leachate in comparison with initial TN in solid samples were approximately 20–23% in S.An.A. reactors and 28.2% in control ones, meaning an overall lower nitrogen emission potential for hybrid lab-scale simulations respect to conventional systems. Stripped ammonia emissions were monitored only in S.An.A.® reactor, in particular in HF (1.9% of initial content in solid samples), enhanced by excess of air circulation and slightly favourable pH conditions (Morello et al., 2016).

Once reached the target values in Table 2, long-term potential emissions of a MSW landfill can still contain slowly degradable organic compounds (almost non-biodegradable with BOD_5/COD < 0.1) with residual concentrations of ammonia nitrogen and sulfates. These compounds are originated by the long-term decomposition of organic matter and can constitute an enduring residual emission, detectable even after the flushing-aeration treatments completion (Ritzkowski and Stegmann, 2013). The evaluation of the environmental relevance of these compounds emissions should be carried out by means of proper application of life cycle assessment or risk assessment procedures (Laner et al., 2012; Brand et al., 2016; Turner et al., 2017).

Total chloride emissions were 90.4–88.6% respect to initial content in solid samples in hybrid reactor and slightly lower in controls (73.8%), depending on solubilisation conditions, pH and leaching processes. Total emission percentages were comparable with data present in literature (Fellner et al., 2009; Raga and Cossu, 2013) and a residual concentration of chlorides will be monitored also during long-term processes.

Heavy metal emissions in leachate are generally not considered to be of environmental interest due to their low mobility (Qu et al., 2008), although they are often present at considerably high levels in solid waste (Oygard et al., 2004). The emission of heavy metals is influenced by pH and other complex chemical equilibriums depending on HM form: ion exchangeable, carbonate-bound (adsorbed), as metal-organic complexes (associated with humic and fulvic acids), as easily reducible metal-oxides, bound to organic matter (other than humic and fulvic acids), in amorphous mineral colloids, as crystalline iron-oxides and bound to aluminum-silicate minerals (Krishnamurti and Naidu, 2002). In this test, total HM emissions did not exceed 10% of initial presence in solid waste, in particular Mn and Ni were more mobile than others, and Fe concentrations remained high until the end of the test (Table 4). A decrease of Mn and Ni emissions was observed in S.An.A.® reactors compared to control ones, while Cu emissions increased of 0.5%.

4.2. Kinetic performances

Reactor performances were mathematically evaluated calibrating the Gomperts model (Eq. (1)) for methane production during Phase 2, the first order kinetic model (Eq. (2)) for COD, ammonia ion depletion during Phase 3, and for chloride solubilization throughout the entire test (Table 5).

As discussed in Cossu et al. (2016), pre-aeration was effective in controlling the concentration of VFA, decreasing pH, stimulating methanogenesis and reducing lag time (λ). In S.An.A.® reactors, total methane potential (BGP) and a higher increase in production kinetics (Rm) was observed than in control columns (Table 5). The duration of Phase 1 was equal to the time required to reach optimum conditions for methanogenesis, while the duration of other phases was calculated as the time required to reach conditions illustrated in Table 2, using the kinetic models calibrated with data from actual reactors.

During forced aeration Phase 3, COD depletion kinetics were generally increased in hybrid reactors, particularly LF. Concomitantly, nitrification-denitrification facilitated ammonia ion depletion, which was consumed 5–10 times kinetically faster than in control reactors, in which leaching effect alone contributed to NH_4^+ attenuation (Table 5).

Chloride leaching kinetics were evaluated in comparison with L/S ratio, not depending directly on biological processes. The results obtained show that 2.19 L/kgTS were sufficient to achieve the target value in anaerobic control reactors, performing better than hybrid columns, which required 2.38–2.58 L/kgTS to achieve the same result (Table 5).

Taking into account all the results obtained, the best performing reactor was LF-I, characterized by low air flow aeration during Phase 3 (7.7 NL/kgTS/d), and by intermittent pre-aeration during Phase 1 (5 NL/kgTS/d, 12 h/d). These conditions were conducive to requiring only 35 days of pre-aeration, a higher production of methane (102.3 NL/kgVS) with faster kinetic (1.7 NL/d/kgVS), and to reaching target conditions illustrated of Table 2 over a period of 105 days, in line with other S.An.A.® reactors.

The total theoretical length of each lab-scale landfill simulation can be estimated as the sum of the theoretical duration of the single phase: which can be evaluated as the time for reaching all the

Table 5
S.An.A.® performance evaluation through calibration of Gompertz model for methane production in Phase 2, first order kinetic model for COD and ammonium ion depletion during Phase 3, and first order kinetic model for chloride leaching throughout the whole test. Duration of single phase was calculated with the calibrated limits model illustrated in Table 2.

Reactor	Phase 1	Phase 2					Phase 3				Entire test		
	Duration (d)	Duration (d)	Gompertz model parameters				COD		N-NH4+		Total duration (d)	Cl−	
			pH	BGP (NL/kgVS)	Rm (NL/d/kgVS)	λ (d)	Duration (d)	Kc (d^{-1})	Duration (d)	Kn (d^{-1})		Total L/S (L/kgTS)	KCl (KgTS/L)
HF-I	55	147	6.46	72.7	1.6	12.3	119	0.010	133	0.025	321	2.58	0.27
HF-C	95	119	6.46	55.5	0.8	15.7	70	0.019	91	0.054	284	2.58	0.31
LF-I	35	166	6.25	102.3	1.7	34.2	91	0.019	105	0.027	292	2.58	0.27
LF-C	81	133	6.34	81.8	1.2	21.5	63	0.028	56	0.041	277	2.38	0.27
ANa	0	270	–	55.8	0.8	168	161	0.011	>203	0.005	431	2.19	0.45
ANb	0	270	–	34.8	0.5	192	161	0.009	>203	0.005	431	2.19	0.45

target concentrations (Table 2) using the kinetics estimated in Table 5. Following this assumption, the total theoretical length of the S.An.A. lab-scale tests amounted to 277–321 d, which is 25–35% lower than the 431 d required by the control reactors. A similar estimation made with landfill water balance, shows that S.An.A. lab-scale tests needed more liquid emissions (L/S = 2.58–2.38 L/kgTS) in comparison with control reactors (2.19 L/kgTS) to enable a drop of the chloride salinity to below target levels. These positive results can be considered after assessing the real time frame of a landfill, which can last for even more than a century (Christensen et al., 2011). For this purpose, some comparative procedure has been developed for anaerobic processes and for old landfill aeration length estimation (Ritzkowski et al., 2016; Hrad et al., 2013). Starting from these concepts, also the S.An.A. technology needs a pilot-scale application to assess the difference between lab and field-scale processes, as well as for evaluating the real magnitude of the potential landfill life reduction.

5. Conclusions

The S.An.A.® bioreactor landfill concept proved to be an efficient system both in increasing methane production and achieving sustainable conditions over a 25–35% lower time frame compared to traditional anaerobic systems.

Pre-aeration was effective in controlling the concentration of VFA, increasing pH and stimulating methanogenesis during the anaerobic phase. S.An.A.® concept proved capable of increasing BGP and kinetics velocity, particularly applying intermittent aeration (5 NL/d/kgTS, 12 h/d).

Forced aeration performed during Phase 3 successfully increased COD depletion kinetics in LF reactors by more than 67%. Moreover, air supply promoted the nitrification process, consuming ammonium ions ten times faster than the leaching effect observed in control columns. By guaranteeing the presence of approx. 4% of oxygen in off gas, reactors with low air flow aeration (7.7 NL/kgTS/d) performed better than those with high air flow, showing faster kinetics, stimulating nitrification and simultaneous efficient denitrification of nitrates produced, with a lower energy consumption.

Chloride leaching was more efficient in control respect to S.An.A.® reactors, reaching target concentrations after L/S = 2.19 L/kgTS, and emitting 73.8% of initial chloride content. Flushing was also the sole means of achieving long-term depletion of ammonium ions and persistent organic compounds in anaerobic control reactors, although less efficient respect to aeration.

The S.An.A.® concept performed best in the presence of intermittent pre-aeration (5 NL/d/kgTS, 12 h/d) during Phase 1, and low air flow forced aeration following methane production during Phase 3.

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References

Agdag, O.N., Sponza, D.T., 2005. Effect of alkalinity on the performance of a simulated landfill bioreactor digesting organic solid wastes. Chemosphere 59, 871–879.
Berge, N.D., Reinhart, D., Dietz, J., Townsend, T., 2006. In situ ammonia removal in bioreactor landfill leachate. Waste Manage. 26, 334–343.
Berge, N.D., Reinhart, D., Batarseh, E.S., 2009. An assessment of bioreactor landfill costs and benefits. Waste Manage. 29, 1558–1567.
Bolyard, S.C., Reinhart, D.R., 2016. Application of landfill treatment approaches for stabilization of municipal solid waste. Waste Manage. 55, 22–30.
Brand, E., De Nijs, T.C.N., Dijkstra, J.J., Comans, R.N.J., 2016. A novel approach in calculating site-specific aftercare completion criteria for landfills in The Netherlands: Policy developments. Waste Manage. 56, 255–261.
CEN, 2002. NEN-EN 12457–4:2002en Compliance test for leaching of granular waste materials and sludges.
Christensen, T.H., Scharff, H., Hjelmar, O., 2011. Landfilling: concepts and challenges. In: Christensen, T.H. (Ed.), Solid Waste Technology and Management, vol. 2. Blackwell Publishing Ltd.
Chugh, S., Clarke, W., Pullammanappallil, P., Rudolph, V., 1998. Effect of recirculated leachate volume on MSW degradation. Waste Manage., 564–573
Cossu, R., Lai., T., Sandon, A., 2012. Standardization of BOD5/COD ratio as a biological stability index for MSW. Waste Manage. J. (32), 1503–1508
Cossu, R., Raga, R., Rossetti, D., 2003. The PAF model: an integrated approach for landfill sustainability. Waste Manage. 23, 37–44.
Cossu, R., Morello, L., Raga, R., Cerminara, G., 2016. Biogas production enhancement using semi-aerobic pre-aeration in a hybrid bioreactor landfill. Waste Manage. 55, 83–92.
Cossu, R., 2016. Back to Earth Sites: From "nasty and unsightly" landfilling to final sink and geological repository. Waste Manage. 55, 1–2.
Delibera Giunta Regionale 7 Oct 2014, no 2461 (D.G.R. 2461/14). Linee guida per la progettazione e gestione sostenibile delle discariche.
EPA, 2015. Bioreactors, Online document 23/04/2015, <http://www.epa.gov/solidwaste/nonhaz/municipal/landfill/bioreactors.htm#1>.
Erses, A.S., Onay, T.T., Yenigun, O., 2008. Comparison of aerobic and anaerobic degradation of municipal solid waste in bioreactor landfill. Biores. Technol. 99, 5418–5426.
European Commission, EU 416/2015. Closing the loop - An EU action plan for the Circular Economy. Brussels, 2.12.2015, no 614.
European Commission, EU 10917/2006. Review of the EU Sustainable Development Strategy. Brussels, 26 June 2006, no 10917.
Fellner, J., Doeberl, G., Allgaier, G., Brunner, P.H., 2009. Comparing field investigations with laboratory models to predict landfill leachate emissions. Waste Manage. 29, 1844–1851.
He, P., Yang, N., Gu, H., Zhang, H., Shao, L., 2011. N_2O and NH_3 emissions from a bioreactor landfill operated under limited aerobic degradation conditions. J. Environ. Sci. 23, 1011–1019.
Heyer, K.U., Hupe, K., Stegmann, R., 2013. Criteria for the Completion of Landfill Aftercare, Sustainable Landfilling. CISA Publisher, pp. 611–622.
Hrad, M., Gamperling, O., Huber-Humer, M., 2013. Comparison between lab- and full-scale applications of in situ aeration of an old landfill and assessment of long-term emission development after completion. Waste Manage. 33, 2061–2073.

Kjeldsen, P., Barlaz, M.A., Rooker, R., Baun, A., Ledin, A., Christensen, T.H., 2002. Present and long-term composition of MSW landfill leachate: a review. Environ. Sci. Technol. 32 (4), 297–336.

Krishnamurti, G.S.R., Naidu, R., 2002. Solid-solution and phytoavailability of copper and zinc in soils. Environ. Sci. Technol. 36, 2645–2651.

Laner, D., Crest, M., Scharff, H., Morris, M.W.F., Barlaz, M.A., 2012. A rewiew of approaches for the long therm management of municipal solid waste landfills. Waste Manage. 32, 498–512.

Lay, J.J., Li, Y.Y., Noike, T., 1998. Developments of bacterial population and methanogenic activity in a laboratory-scale landfill bioreactor. Water Resour. 32, 3673–3679.

Lornage, R., Redon, E., Lagier, T., 2013. Carbon and nitrogen mass balance for different landfill concepts, Lysimeter tests. In: Cossu, R., Van der Sloot, H.A. (Eds.), Sustainable Landfilling. CISA Publisher, pp. 95–105.

Morello, L., Cossu, R., Raga, R., Pivato, A., Lavagnolo, M.C., 2016. Recirculation of reverse osmosis concentrate in lab-scale anaerobic and aerobic landfill simulation reactors. Waste Manage. 56, 262–270.

Norbu, T., Visvanathan, C., Basnayake, B., 2005. Pretreatment of municipal solid waste prior to landfilling. Waste Manage. 25, 997–1003.

Oygard, K.J., Mage, A., Gjengedal, E., 2004. Estimation of the mass-balance of selected metals in four sanitary landfills in Western Norway, with emphasis on the heavy metal content of the deposited waste and the leachate. Water Res. J. 2004, 2851–2858.

Price, G.A., Barlaz, M.A., Hater, G.R., 2003. Nitrogen management in bioreactor landfills. Waste Manage. 23, 675–688.

Qu, X., He, P.J., Shao, L.M., Lee, D.J., 2008. Heavy metals mobility in full-scale bioreactor landfill: Initial stage. Chemosphere 70, 769–777.

Raga, R., Cossu, R., 2013. Bioreactor tests preliminary to landfill in situ aeration: A case study. Waste Manage. 33, 871–880.

Raga, R., Cossu, R., Heerenklage, J., Pivato, A., Ritzkowski, M., 2015. Landfill aeration for emission control before and during landfill mining. Waste Manage. 46, 420–429.

Repetti, R., Testolin, G., Cossu, R., Raga, R., 2013. "Discarica S.AN.A® ": Un sistema innovativo per la sostenibilità ambientale delle discariche. In: "Proceedings Sardinia 2013". Fourteenth International Waste Management and Landfill Symposium, S. Margherita di Pula, Cagliari, Italy, 30 September – 4 October 2013.

Ritzkowski, M., Heyer, K.-U., Stegmann, R., 2006. Fundamental processes and implications during in situ aeration of old landfills. Waste Manage. 26, 356–372.

Ritzkowski, M., Stegmann, R., 2013. Landfill aeration within the scope of post-closure care and its completion. Waste Manage. 33, 2074–2082.

Ritzkowski, M., Walker, B., Kuchta, K., Raga, R., Stegmann, R., 2016. Aeration of the Teuftal landfill: Field scale concept and lab scale simulation. Waste Manage. 55, 99–107.

Sandip, M., Kanchan, K., Ashok, B., 2012. Enhancement of methane production and bio-stabilisation of municipal solid waste in anaerobic biorector landfill. Biores. Technol. 110, 10–17.

Sang, N.N., Soda, S., Inoue, D., Sei, K., Ike, Michihiko, 2009. Effects of intermittent and continuous aeration on accelerative stabilization and microbial population dynamics in landfill bioreactors. J. Biosci. Bioeng. 108, 336–343.

Sekman, E., Top, S., Varank, G., Bilgili, M.S., 2011. Pilot-scale investigation of aeration rate effect on leachate characteristics in landfills. Fresenius Environ. Bull. 20.

Shao, L.-M., He, P.-J., Li, G.-J., 2008. In situ nitrogen removal from leachate bioreactor landfill with limited aeration. Waste Manage. 20, 1000–1007.

Sun, F., Wang, Y.N., Sun, X., Wu, H., Zhang, H., 2013. Production characteristics of N_2O during stabilization of municipal solid waste in an intermittent aerated semi-aerobic bioreactor landfill. Waste Manage. 33, 2729–2736.

Townsend, T.G., Powell, J., Jain, P., Xu, Q., Tolaymat, T., Reinhart, D., 2015. Sustainable Practices for Landfill Design and Operation. Springer, Series: Waste Management Principles and Practice.

Turner, D.A., Beaven, R.P., Woodman, N.D., 2017. Evaluating landfill aftercare strategies: A life cycle assessment approach. Waste Manage. 63, 417–431.

Wang, Q., Kuninobu, M., Kakimoto, K., Ogawa, H.I., 1999. Upgrading of anaerobic digestion of waste activated sludge by ultrasonic pretreatment. Biores. Technol. 68, 309–313.

Wu, C., Shimaoka, T., Nakayama, H., Komiya, T., Chai, X., Hao, Y., 2014. Influence of aeration modes on leachate characteristic of landfills that adopt the aerobic–anaerobic landfill method. Waste Manage. J. 34 (2014), 101–111.

Xu, Q., Jin, Z., Ma, Z., Tao, H., Ko, J.H., 2014. Methane production in simulated hybrid bioreactor landfill. Biores. Technol. 168, 92–96.